Project: Worked examples
Title: Shaft: static, fatigue, keys, critical speed
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A stepped (optionally hollow) shaft on two bearings carrying radial loads in two planes and a steady torque: bending moment, slope and deflection along it, the fatigue and yield safety factors at the sections the user marks, slopes at the bearings, deflection at the loads, the first lateral critical speed and parallel keys.
Bending: the shaft as an S2 beam (pin at A, roller at B) in each plane; M = √(M_y² + M_z²) Stress at a section: σ_a = K_f·32M_a·d/(π(d⁴ − d_i⁴)), τ_m = K_fs·16T_m·d/(π(d⁴ − d_i⁴)); K_f = 1 + q(K_t − 1) Endurance: S_e = k_a·k_b·k_c·k_d·k_e·k_f·S_e′, S_e′ = 0.5·S_ut (S_ut ≤ 1400 MPa, else 700 MPa); k_a = a·S_ut^b, k_b = 1.24d^−0.107 or 1.51d^−0.157, k_e = 1 − 0.08·z_R Fatigue (DE-Goodman): 1/n_f = σ_a/S_e + √3·τ_m/S_ut; yield n_y = S_y/√((σ_a)² + 3τ_m²) Critical speed: Rayleigh ω = √(g·Σm·y/Σm·y²); Dunkerley 1/ω² = Σ m_i·δ_ii Parallel key: τ = 2T/(d·w·L), σ = 4T/(d·h·L)
Ref: Shigley's Mechanical Engineering Design 11th ed., Ch. 6 (§6-9 Marin factors), Ch. 7 (§7-4 to §7-7) — Fatigue, shaft stresses, deflection, critical speeds, keys
Ref: Mott, Machine Elements in Mechanical Design 6th ed., Ch. 12 — Shaft design

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segments	Shaft segments (left to right)	5 row(s)		drawing: GBX-SH-01 rev C	any
segments[1]		x_end = 40 mm; d = 35 mm; d_i = 0 mm			
segments[2]		x_end = 120 mm; d = 40 mm; d_i = 0 mm			
segments[3]		x_end = 180 mm; d = 45 mm; d_i = 0 mm			
segments[4]		x_end = 260 mm; d = 40 mm; d_i = 0 mm			
segments[5]		x_end = 340 mm; d = 35 mm; d_i = 0 mm			
x_A	Bearing A position	20	mm	drawing: GBX-SH-01 rev C	0 ... 3000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
x_B	Bearing B position	280	mm	drawing: GBX-SH-01 rev C	1 … 3000 mm
loads	Radial loads and masses	1 row(s)		spec: Gear mesh forces from the gear calculation	any
loads[1]		x = 150 mm; F_y = -2.2 kN; F_z = 6 kN; m = 4 kg			
T	Torque transmitted	0.3	kN*m	spec: Motor rated torque × service factor (E1)	0 … 50 kN*m
x_T1	Torque enters at	150	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
x_T2	Torque leaves at	330	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
n	Operating speed	1450	rpm	spec: Motor nameplate	0 … 30000 rpm
sections	Sections to check	4 row(s)		assumption: Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_ts 3.0); q from the Neuber fit.	any
sections[1]		x = 120 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9			
sections[2]		x = 150 mm; K_t = 2.14; K_ts = 3; q = 0.8; q_s = 0.9			
sections[3]		x = 180 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
sections[4]		x = 260 mm; K_t = 1.6; K_ts = 1.4; q = 0.8; q_s = 0.9			
S_ut	Ultimate tensile strength	700	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	300 … 2000 MPa
S_y	Yield strength	500	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	150 … 1900 MPa
E	Young's modulus	207	GPa	assumption: Steel.	50 … 250 GPa
rho	Density (shaft self-mass in the critical speed)	7850	kg/m^3	assumption: Steel.	0 … 20000 kg/m^3
finish	Surface finish	machined or cold-drawn		drawing: GBX-SH-01 rev C	any
R	Reliability	99	%	assumption: Design reliability.	50 … 99.9 %
k_d	Temperature factor k_d	1		assumption: Below 250 °C.	0.5 … 1.1
k_f	Miscellaneous factor k_f	1		assumption: No other effects.	0.3 … 1.5
n_target	Target safety factor	1.5		spec: Company design standard	1.2 … 4
theta_max	Slope limit at the bearings	0.001	rad	assumption: Deep-groove ball bearing slope limit.	0.00001 … 0.1 rad
y_max	Deflection limit at the loads	0.13	mm	assumption: Spur gear deflection limit.	0.001 … 10 mm
x_key	Key position	150	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
w	Key width	14	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
h	Key height	9	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
L_key	Key length	40	mm	drawing: GBX-SH-01 rev C	1 … 500 mm
S_y_key	Key yield strength	400	MPa	spec: C45 key steel	150 … 1500 MPa

3. Assumptions

A1. Linear elastic, two simple bearing supports (pin at A, roller at B); bending fully reversed by rotation, torque steady. (Shigley 11th ed. §7-4.)
A2. No torsional vibration; only the first lateral mode, by Rayleigh with Dunkerley as a lower bound. (Shigley 11th ed. §7-6.)
A3. K_t and K_ts come from user records or published fit equations; no stress-concentration chart is reproduced. (§A5.)
A4. At a shoulder the smaller diameter governs; k_c = 1 because the stresses are combined by von Mises. (Shigley 11th ed. §6-9, §7-4.)
A5. sections (Sections to check) is taken as 4 row(s). (Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_ts 3.0); q from the Neuber fit.)
A6. E (Young's modulus) is taken as 207 GPa. (Steel.)
A7. rho (Density (shaft self-mass in the critical speed)) is taken as 7850 kg/m^3. (Steel.)
A8. R (Reliability) is taken as 99 %. (Design reliability.)
A9. k_d (Temperature factor k_d) is taken as 1. (Below 250 °C.)
A10. k_f (Miscellaneous factor k_f) is taken as 1. (No other effects.)
A11. theta_max (Slope limit at the bearings) is taken as 0.001 rad. (Deep-groove ball bearing slope limit.)
A12. y_max (Deflection limit at the loads) is taken as 0.13 mm. (Spur gear deflection limit.)

4. Working

[17] R_A = √(R_A,y² + R_A,z²)
= √(1.1² + -3²)
= 3.1953 kN
Radial reaction at bearing A

[18] R_B = √(R_B,y² + R_B,z²)
= √(1.1² + -3²)
= 3.1953 kN
Radial reaction at bearing B

[19] d_A = seat diameter at A
= x = 20 mm
= 35 mm
Shaft diameter at bearing A

[20] d_B = seat diameter at B
= x = 280 mm
= 35 mm
Shaft diameter at bearing B

[21] M_max = max √(M_y² + M_z²) along the shaft
= 37 stations
= 0.41539 kN·m
Largest bending moment

[22] theta_A = slope at A (two planes)
= x = 20 mm
= 0.0008961 rad
Slope at bearing A

[23] theta_B = slope at B (two planes)
= x = 280 mm
= 0.0008961 rad
Slope at bearing B

[24] y_load = largest deflection at a load
= 0.071775 mm at 150
= 0.071775 mm
Deflection at the loads

[26] k_a = a·S_ut^b — Shigley 11th ed. §6-9
= 4.51 × 700^-0.265 (machined or cold-drawn)
= 0.79474

Surface factor

[27] k_e = 1 − 0.08·z_R — Shigley 11th ed. §6-9

= 1 − 0.08 × 2.3263

= 0.81389

Reliability factor

[28] S_e1 = 0.5·S_ut

= 700 MPa

= 350 MPa

Rotating-beam endurance limit

[29] X1 = σ_a = K_f·32M/(πd³'), σ_m = K_fs·16T/(πd³'); 1/n_f = σ_a/S_e + σ_m/S_ut — Shigley 11th ed. §6-8

= x = 120 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_fs = 1.45, M = 319.53 N·m, σ_a = 79.334 MPa, σ_m = 0 MPa, n_y = 6.3025

= 2.3845

Section at 120 mm: fatigue safety factor (DE-Goodman)

[30] X2 = σ_a = K_f·32M/(πd³'), σ_m = K_fs·16T/(πd³'); 1/n_f = σ_a/S_e + σ_m/S_ut — Shigley 11th ed. §6-8

= x = 150 mm, d = 45 mm, k_b = 0.82514, S_e = 186.8 MPa, K_f = 1.912, K_fs = 2.8, M = 415.39 300 N·m, σ_a = 88.778 MPa, σ_m = 46.947 MPa, n_y = 4.1532

= 1.6909

Section at 150 mm: fatigue safety factor (DE-Goodman)

[31] X3 = σ_a = K_f·32M/(πd³'), σ_m = K_fs·16T/(πd³'); 1/n_f = σ_a/S_e + σ_m/S_ut — Shigley 11th ed. §6-8

= x = 180 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_fs = 1.45, M = 319.53 300 N·m, σ_a = 79.334 MPa, σ_m = 34.616 MPa, n_y = 5.0281

= 1.9801

Section at 180 mm: fatigue safety factor (DE-Goodman)

[32] X4 = σ_a = K_f·32M/(πd³'), σ_m = K_fs·16T/(πd³'); 1/n_f = σ_a/S_e + σ_m/S_ut — Shigley 11th ed. §6-8

= x = 260 mm, d = 35 mm, k_b = 0.84763, S_e = 191.9 MPa, K_f = 1.48, K_fs = 1.36, M = 63.906 300 N·m, σ_a = 22.47 MPa, σ_m = 48.465 MPa, n_y = 5.7538

= 4.2192

Section at 260 mm: fatigue safety factor (DE-Goodman)

[33] S_e = k_a·k_b·k_c·k_d·k_e·k_f·S_e' (governing section)

= k_c = 1

= 186.8 MPa

Endurance limit at the governing section

[34] n_f_min = min over sections

= 4 section(s)

= 1.6909

Lowest fatigue safety factor

[35] n_y_min = min over sections S_y/σ'_max

= 4 section(s)

= 4.1532

Lowest yield safety factor

[36] n_c = √(g·δm·y/δm·y²) (Rayleigh) — Shigley 11th ed. §7-6

= δm = 7.2337 kg over 37 nodes

= 39515 rpm

First lateral critical speed (Rayleigh)

[37] n_c_D = 1/δ² = √ m_i·δ_ii (Dunkerley, lower bound) — Shigley 11th ed. §7-6

= 37 masses

= 37295 rpm

First critical speed, Dunkerley lower bound

[38] r_c = n_c/n

= 39515 / 1450

= 27.251

Critical-speed ratio

[43] tau_key = 2T/(d·w·L) Shigley 11th ed. §7-7
= 2 × 300 / (45 × 14 × 40) mm³
= 23.81 MPa
Key shear stress
[44] sigma_key = 4T/(d·h·L) Shigley 11th ed. §7-7
= 4 × 300 / (45 × 9 × 40) mm³
= 74.074 MPa
Key bearing stress
[46] L_key_req = max(2T·n/(d·w·0.577S_y), 4T·n/(d·h·S_y))
= n = 1.5
= 11.111 mm
Key length required
[47] u_key = L_req/L
= 11.111 / 40
= 0.27778
Key utilisation
[48] u_f = n_target/n_f,min
= 1.5 / 1.6909
= 0.88712
Fatigue utilisation
[49] u_y = n_target/n_y,min
= 1.5 / 4.1532
= 0.36117
Yield utilisation
[50] u_slope = max(□_A, □_B)/□_max
= 0.0008961 / 0.001
= 0.8961
Slope utilisation
[51] u_defl = y_load/y_max
= 0.071775 / 0.13 mm
= 0.55212
Deflection utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	3.1953 kN	Reaction at A
R_B	3.1953 kN	Reaction at B
d_A	35 mm	Seat diameter at A
d_B	35 mm	Seat diameter at B
M_max	0.41539 kN·m	Largest bending moment
theta_A	0.0008961 rad	Slope at A
theta_B	0.0008961 rad	Slope at B
y_load	0.071775 mm	Deflection at the loads
S_e	186.8 MPa	Endurance limit
n_f_min	1.6909	Lowest fatigue safety factor
n_y_min	4.1532	Lowest yield safety factor
n_c	39515 rpm	First critical speed (Rayleigh)

SYMBOL	VALUE	DESCRIPTION
n_c_D	37295 rpm	Dunkerley lower bound
r_c	27.251	Critical-speed ratio
tau_key	23.81 MPa	Key shear
sigma_key	74.074 MPa	Key bearing
L_key_req	11.111 mm	Key length required
u_key	0.27778	Key utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Slope at the bearings against limit	u_slope = 0.8961	u_slope ≤ 1	0.896	
PASS	Fatigue safety factor against target (DE-Goodman)	u_f = 0.88712	u_f ≤ 1	0.887	Shigley 11th ed. §7-4
PASS	Deflection at the loads against limit	u_defl = 0.55212	u_defl ≤ 1	0.552	
PASS	Yield safety factor against target	u_y = 0.36117	u_y ≤ 1	0.361	
PASS	Key length against required	u_key = 0.27778	u_key ≤ 1	0.278	

7. Applicability

WARNING — record.S_ut: S_ut: a tabulated value not taken from a reference record (Shaft material); enter it as one so its source, edition and licence travel with it.
WARNING — record.S_y: S_y: a tabulated value not taken from a reference record (Shaft material); enter it as one so its source, edition and licence travel with it.
WARNING — record.E: E: a tabulated value not taken from a reference record (Shaft material); enter it as one so its source, edition and licence travel with it.
WARNING — record.theta_max: theta_max: a tabulated value not taken from a reference record (Slope and deflection limits); enter it as one so its source, edition and licence travel with it.
WARNING — record.y_max: y_max: a tabulated value not taken from a reference record (Slope and deflection limits); enter it as one so its source, edition and licence travel with it.

8. Verification

Template mech.shaft@1.0.0 Content hash 9955fd0fa27243ad Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0